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Enhancement of piezoelectric properties of (LiCePr)-multidoped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ high temperature ceramics



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ABSTRACT

Aurivillius type (LiCePr)-multidoped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics were prepared using the conventional mixed oxides route. The effects of the (LiCePr) additives on the structure and dielectric properties were studied. The piezoelectricity was significantly improved by the incorporation of (LiCePr) dopants. The $\text{Ca}_{0.9}(\text{Li}_{0.5}\text{Ce}_{0.25}\text{Pr}_{0.25})_{0.1}\text{Bi}_2\text{Nb}_2\text{O}_9$ (CBN–LCP10) ceramics possess the optimum properties, and the d_{33} , k_p and T_C values were found to be ~ 17.3 pC/N, 7.8% and ~ 939 °C, respectively. The thermal depoling behavior and electromechanical properties at elevated temperature were studied in detail. Both results demonstrate that CBN–LCP10 ceramics possess a stable piezoelectricity in a wide temperature range, making CBN-based ceramics promising for high temperature applications.

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1. Introduction

Environment friendly perovskite structure piezoelectric ceramics have been intensively studied in recent years [1,2]. However, the low Curie temperature limits their applications in some hazardous environments. Aurivillius-type bismuth layer-structured ferroelectrics have attracted much attention for their potential use in high-temperature piezoelectric devices, owing to their high Curie temperature ($T_C > 500$ °C) [3,4].

$\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CBN) material, a typical two-layered Aurivillius type compound, possesses the known high Curie temperature (~ 930 °C) and thermal depoling temperature for polycrystalline ceramics [5,6]. However, the piezoelectric coefficient d_{33} of pure CBN ceramic is relatively low (~ 7 pC/N). Many investigations have been carried out to improve the piezoelectric properties including texturing techniques and/or chemical modifications. It was an effective way to improve the piezoelectric properties using rare-earth elements La^{3+} and/or Ce^{3+} dopants but degraded the T_C value [7–9]. The d_{33} value was significantly improved to be ~ 17.8 pC/N via grain orientation techniques, and the T_C value was not affected [6]. On the other hand, it has been reported that photoluminescence property of Pr^{3+} ions doped $\text{CaBi}_2\text{Ta}_2\text{O}_9$ ceramics in recent years. Unfortunately, piezoelectric properties were not mentioned [10]. Moreover, two or more kinds of rare-earth elements codoped CBN ceramics were rarely reported. In conclusion, it is of great interest to study the effects of piezoelectric

properties of two or more kinds of rare-earth elements doped CBN compounds.

In the present paper, we focus on the effects of (LiCePr)-multimodified CBN ceramics prepared by the mixed oxide routes. The crystal structure, dielectric behavior and piezoelectric properties were investigated. Furthermore, the thermal depoling behavior and electrical impedance modulus at elevated temperatures were also addressed.

2. Material and methods

$\text{Ca}_{1-x}(\text{Li}_{0.5}\text{Ce}_{0.25}\text{Pr}_{0.25})_x\text{Bi}_2\text{Nb}_2\text{O}_9$ (abbreviated as CBN–LCP100x, $x=0, 0.02, 0.04, 0.06, 0.08$ and 0.10) ceramics were prepared by standard ceramic preparation routes. Bi_2O_3 (99%), CaCO_3 (99.8%), Nb_2O_5 (99.9%), Li_2CO_3 (99.99%), Pr_2O_3 (99.99%) and CeO_2 (99.99%) powders were used as raw materials. All powders were weighed according to the stoichiometric compositions and subsequently ball-mixed for 24 h. Calcination was carried out at ~ 900 °C for 3 h, and then these mixtures were milled again in the same condition. The powders were pressed into pellets followed by a cold-isostatic pressing under 200 MPa for 10 min using polyvinyl alcohol (PVA) as a binder. After burning out the PVA at ~ 550 °C, these pellets were sintered at 1120 – 1150 °C for 2 h depending on the x value. The phase structure of these ceramics was determined by X-ray diffraction (XRD) employing the $\text{Cu K}\alpha$ lines (DX2700, Dandong, China). For the electrical measurements, Pt electrodes were fired on both surfaces of these sintered pellets at 1000 °C for 10 min in air. The temperature behavior as a function of temperature was determined using a precision LCR analyzer (HP4980A, Agilent, US) connecting with a programmable furnace. Samples were poled in a

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silicon oil bath at 180 °C under a DC electric field of 10–14 kV/mm for 15 min. The piezoelectric coefficient d_{33} was measured using a quasi-static d_{33} meter (ZJ-3A, Institute of Acoustics, Academia Sinica, China). The electrical impedance modulus was determined by the precision impedance analyzer (HP4294A, Agilent, US). Thermal depoling experiments were conducted by holding the poled samples for 2 h at each temperature, cooling to room temperature, measuring the d_{33} value, and repeating the procedures up to 950 °C.

3. Results and discussion

Fig. 1 shows the XRD patterns performed on the pure or (LiCePr)-substituted CBN ceramics. The diffraction peaks were indexed according to the JCPDS 49-0608 ($\text{CaBi}_2\text{Nb}_2\text{O}_9$) and shown in Fig. 1. All the samples have a single phase of two layered Aurivillius type structure and no secondary phases are found, indicating that (LiCePr) dopants have diffused into the CBN lattice to form a solid solution. It can be seen that the main peaks shift to higher angles with the incorporation of (LiCePr) dopants (Fig. 1 (b)). Furthermore, the splitting of (200) and (020) peaks and shifting of 2θ angles imply that (LiCePr) dopants could induce the lattice distortion [11,12]. This phenomenon is attributed to the partial substitution voluminous Ca^{2+} (~ 1.34 Å) by Li^+ (~ 0.92 Å), Ce^{3+} (~ 1.34 Å) and Pr^{3+} (~ 1.14 Å) ions, decreasing the lattice parameters [13,14]. However, with a further increase of x value up to 0.10, the diffraction peaks shift to lower angles arising from a possible chance that Ce^{4+} ions occupy the B-sites.

Temperature dependence of (a) permittivity and (b) loss tangent of CBN-LCPx ceramics measured at 1 MHz were shown in Fig. 2. The Curie-temperature of pure CBN ceramics is ~ 930 °C, which is in agreement with other reports [6,9]. It is well known that an increase of lattice distortion leads to a higher T_C value and *vice versa* in Aurivillius type materials [15,16]. As expected, the T_C value shifts slightly to a higher temperature (~ 945 °C) with $x=0.06$, but decrease with a further increase of x concentration (see the inset of Fig. 2). Pure or modified CBN-based ceramics show high T_C values. This provides the potential applications in ultra-high temperature environments. Besides, it is noted that the variation of permittivity and loss values below 650 °C are relatively flat indicating that CBN-based ceramics possess a stable dielectric property.

Fig. 3(a) shows the d_{33} values as a function of the x value measured at room temperature. It can be clearly seen that the d_{33} value increased with an increase of (LiCePr) content. The d_{33} value of CBN-LCP10 ceramics is obtained to be 17.3 pC/N. This high level

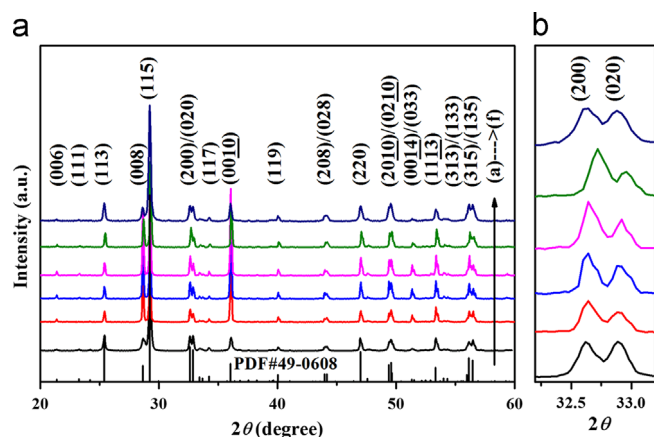


Fig. 1. (a) XRD patterns and (b) expanded XRD patterns of CBN-LCP100x sintered ceramics (a) $x=0$, (b) $x=0.02$, (c) $x=0.04$, (d) $x=0.06$, (e) $x=0.08$ and (f) $x=0.10$.

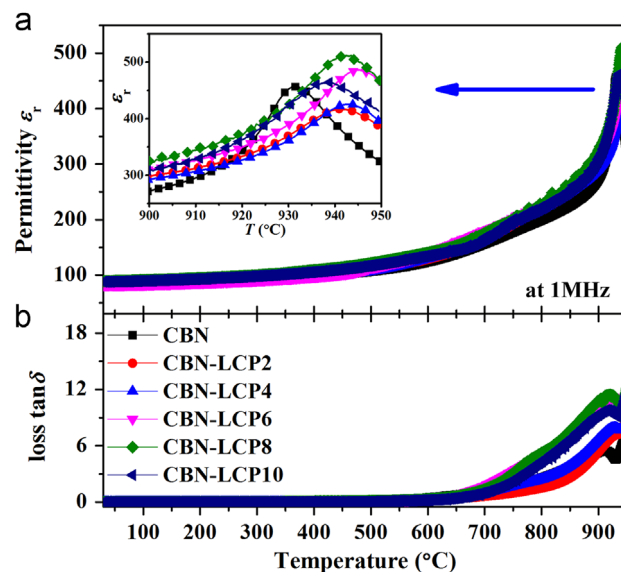


Fig. 2. Temperature dependence of (a) permittivity and (b) dielectric loss of CBN-LCP100x ceramics.

d_{33} value is nearly three times of that of pure-CBN ceramics (~ 5.8 pC/N) and is comparable with that of the texture sample (~ 17.8 pC/N) [6]. The incorporation of (LiCePr) dopants increases the “chaos or rattling space” in local perovskite structure then facilitates the movement of domain walls and thus significantly enhanced the d_{33} value [17,18].

The thermal depoling behavior of pure or modified CBN ceramics is shown in the Fig. 3(b). The d_{33} value of all compounds decreases rapidly to zero when the annealing temperature is above the T_C . This exhibits the nature of the ferro-paraelectric phase transition. The d_{33} value decreases slightly at low temperature regions due to the thermal activated reverse switching of the non-180° domains [7,19]. Even annealing at 850 °C for 2 h, the d_{33} value of CBN-LCP10 ceramics retains ~ 15.3 pC/N (88% of its initial values). This result demonstrates that the CBN-LCP100x ceramics possess stable piezoelectric properties at various temperatures.

The resonance and anti-resonance method is invoked to describe the real-time behavior of electromechanical coupling properties of piezoelectric ceramics at high temperatures [20]. Fig. 3(c) depicts the electrical impedance modulus of CBN-LCP10 ceramics as a function of frequency at various temperatures. And the resonance frequency (f_r) and planar electromechanical coupling factor (k_p) of CBN-LCP10 ceramics at selected temperature are shown in Fig. 3(d). It is noted that the f_r value is in the vicinity of 300 kHz, and shifts to lower frequency, showing the negative temperature coefficient behavior. Moreover, the CBN-LCP10 composition has a low temperature coefficient of resonance frequency ($\text{TK}f_r$) of -54 ppm/K in a wide temperature region (RT–700 °C) [21]. The amplitude of impedance modulus decreases with rising temperatures up to 700 °C, indicating that the material becomes softer. Furthermore, the k_p value is found to be 7.8% at room temperature, and remains on the order of 8% over the temperature range from room temperature to 700 °C, exhibiting very stable piezoelectric properties. The impedance modulus results confirm that piezoelectric properties of CBN-LCP10 composition do exist in the measurement temperature range.

4. Conclusions

In summary, the Aurivillius type $\text{Ca}_{1-x}(\text{Li}_{0.5}\text{Ce}_{0.25}\text{Pr}_{0.25})_x\text{Bi}_2\text{Nb}_2\text{O}_9$ (CBN-LCP100x) ceramics were prepared via conventional

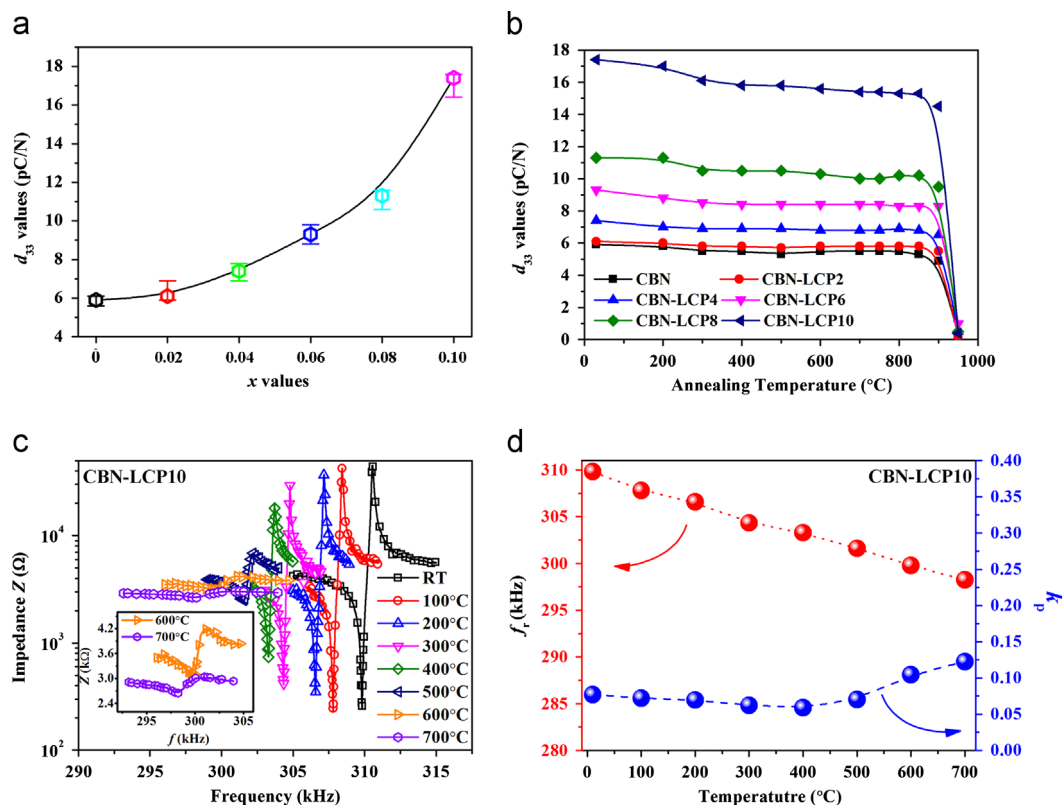


Fig. 3. (a) d_{33} values at room temperature vs. the x content; (b) thermal depoling behavior of CBN-LCP100x ceramics; (c) electrical impedance modulus as a function of frequency for CBN-LCP10 ceramics at selected temperatures; (d) resonance frequency f_r and electromechanical coupling factor k_p at various temperatures for CBN-LCP10 ceramics.

mixed oxide routes. The crystal structure of CBN-LCP100x ceramics was the presence of a single two-layered Aurivillius type structure. This work also presented the considerable influence of (LiCePr) dopants on the dielectric and piezoelectric properties. The Curie temperature, T_C value of modified specimens shifted to higher temperatures than that of pure CBN ceramics. The piezoelectric coefficients d_{33} value was dramatically increased by the (LiCePr) modifications. The $\text{Ca}_{0.9}(\text{Li}_{0.5}\text{Ce}_{0.25}\text{Pr}_{0.25})_{0.1}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics showed optimum piezoelectric properties, and the d_{33} , k_p , T_C values were found to be ~ 17.3 pC/N, $\sim 7.8\%$ and ~ 939 °C, respectively. In addition, it was demonstrated that the CBN-LCP10 ceramics possess the stable piezoelectric properties via the thermal depoling behavior and electrical impedance modulus at elevated temperature.

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